

Computer Simulation

Lesson 10: ALOHA

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Special thanks to: Dr. Pham Thi Huong

ALOHA

- Introduction
 - What is ALOHA network
 - Pure ALOHA and slotted ALOHA
- Performance evaluation
 - Predict network throughput

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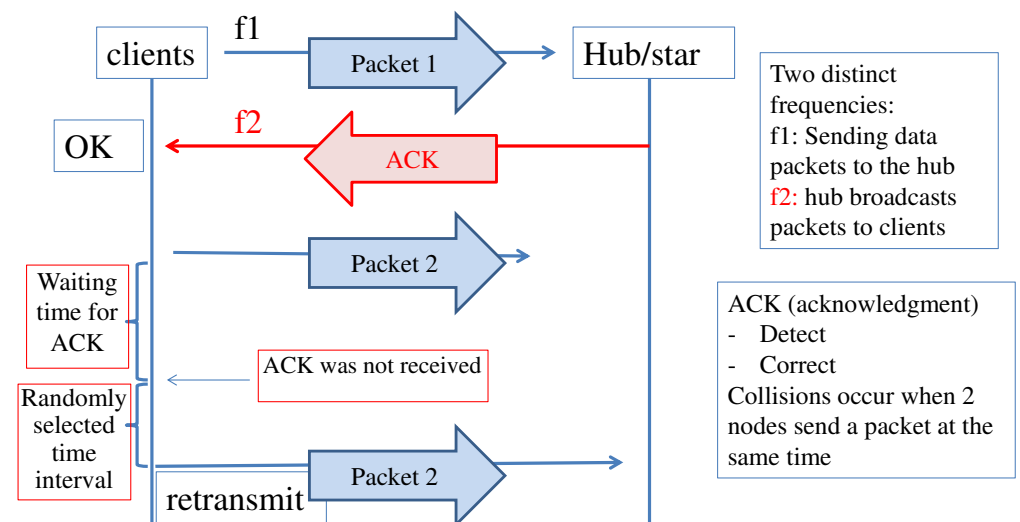
Introduction to ALOHA

- ALOHA: Additive Links On-line Hawaii Area
- A pioneering computer networking system developed at the University of Hawaii
- The first public demonstration of a wireless packet data network in June, 1971
- ALOHA is used as a protocol in the second layer of the OSI model for LAN and broadcast protocol

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The original version of ALOHA



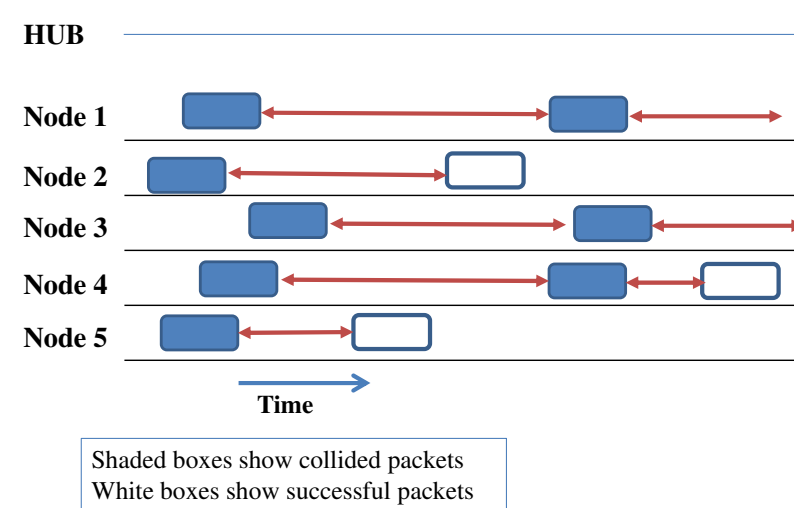
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Pure ALOHA (1/2)

- All nodes communicate with the network on the same frequency
- ACK scheme is used to deal with collisions
- Pure ALOHA or random-access channel
 - If a node has data → send data immediately
 - If a node sends data while another node is transmitting data → collision will occur → data will be resent after a random back-off time
- Advantages
 - Reduce the complexity of the protocol and the networking hardware (nodes do not need negotiate who is allowed to send data)
- Disadvantages
 - Data is lost because of collision
 - Time is wasted

Pure ALOHA (2/2)



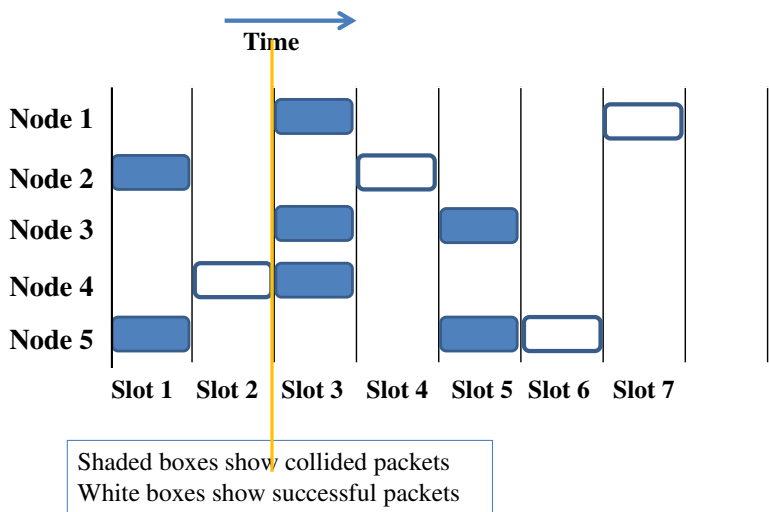
Question for pure ALOHA

- How to reduce collisions
- Need a mechanism to control: which node can send data and when it is sent

Slotted ALOHA (1/2)

- Time is divided into “time slots”: one time slot is one packet duration
- A packet is only sent in the start of a time slot
→ When a node wants to send packet → wait for the next slot
- Require synchronization → to inform to each node when a time slot begins
- Advantages
 - Reduce the number of collisions

Slotted ALOHA (2/2)



Performance Evaluation

- To assess a pure or slotted ALOHA, throughput is an important indicator **Goodput**
- Throughput or network throughput is the rate of successful message delivery over a communication channel (saturation throughput)
- A transmission is successful if only one single packet is present in the slot
- If maximum throughput of the ALOHA is T per a unit of time $\rightarrow 100(1-T) \%$ channel capacity will be lost because of collision

Saturation throughput

- Calculate throughput of a slotted ALOHA
- Access probability of user i is x_i with $i \in \{1, \dots, n\}$
- Saturation throughput of user i is given by

$$S_i = x_i \prod_{j=1, j \neq i}^n (1 - x_j)$$

Example

$$x_1 = 0.1, x_2 = 0.2, x_3 = 0.3$$

$$S_1 = x_1 \times (1 - x_2) \times (1 - x_3)$$

$$S_2 = x_2 \times (1 - x_1) \times (1 - x_3)$$

$$S_3 = x_3 \times (1 - x_1) \times (1 - x_2)$$

Node 1	Transmit	Not transmit	Not transmit
Node 2	Not transmit	Transmit	Not transmit
Node 3	Not transmit	Not transmit	Transmit
Result	Node 1 successful	Node 2 successful	Node 3 successful

Practice 2: Slotted ALOHA

- A user arrives to connect to the network according to a Poisson process with the rate of 10 / hour
- Each user stays at the place for a time which is following an exponential distribution with the average of 0.5 hour
- Access probability of each user is p
 $0.1 \leq p \leq 0.9$
- Simulation length: 24 hours
- Length of a time-slot: 0.01 hour

Assignment

- Calculate the average throughput of existing users as a function of time when $p = 0.3$
- Analyze how the average throughput changes according to p ($0.1 \leq p \leq 0.9$, step size=0.1)
- Generate a figure

Submission

- Divide into several teams and discuss
 - New teams launched
- **3rd July**: Each team will make a presentation to introduce the solution (report, MATLAB program)
- Each student will grade
 - The solutions of his / her team and other teams
 - The contribution of his / her team members

Summary

- Reference
 - <http://ktmt.github.io/blog/2013/11/03/aloha-protocol/>
 - <https://en.wikipedia.org/wiki/ALOHAnet>
 - Computer Simulation course 2016, Bo Gu